

The Zeta Function of the T^4/Z_3 Torus:

Orbifold Projection, Dirichlet L-Functions and
Sector Pairing of the $GF(27)$ Classes

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The Zeta Function of the T^4/Z_3 Torus

Abstract

The Galois factorisation classes of Doc. 342 and the harmonic series on the T^4 torus of Doc. 159 together yield a precise statement about the zeta function of the FFGFT spacetime.

Four results: **(A)** The Z_3 orbifold projection of T^4/Z_3 transforms the full Riemann zeta into a Dirichlet L-function: $\zeta_{T^4/Z_3}(s) = L(s, \chi_0) = (1 - 3^{-s})\zeta(s)$ **[B]**. **(B)** The Euler product of this L-function is hierarchically ordered by the entry depth $k = \text{ord}_p(3)$; the physically visible primes $\{2, 3, 5, 7, 11, 13\}$ are exactly those with $k \leq 6$ **[B]**. **(C)** The orbifold projection generates additional zeros at $s = 2\pi i k / \ln 3$ ($k \in \mathbb{Z} \setminus \{0\}$) on the imaginary axis $\text{Re}(s) = 0$; these are the algebraic signature of the Z_3 sector separation **[B]**. **(D)** The four Frobenius conjugacy classes of primitive elements in $\text{GF}(27)^*$ (Doc. 342, Theorem A) are paired into two classes by the FFGFT sector pairing $k \leftrightarrow -k$ (Doc. 336; on $\text{GF}(27)^*$: $x \mapsto x^{-1}$), $f_1 \leftrightarrow f_4$ and $f_2 \leftrightarrow f_3$; negation $x \mapsto -x = x^{13}x$ couples each primitive class to exactly one order-13 class **[B]**. The Galois character of $\text{Gal}(\text{GF}(27)/\text{GF}(3)) \cong Z_3$ does not distinguish the classes; the distinction lies in $(\mathbb{Z}/26)^*/\langle 3 \rangle \cong Z_4$. Whether exactly one second $\text{GF}(27)^*$ layer besides the lepton layer is physically realised remains open **[S]**.

Check scripts: `pruef_343_zeta_galois.py`, `pruef_343b_klassen_symmetrie.py`.

.1 Status markers and starting point

Status markers

- **[K]** — *Confirmed*: fully proved from ξ or a corpus-established derivation; all check-script assertions passed.
- **[B]** — *Established*: numerically secured; analytical proof present or trivial.
- **[S]** — *Speculative*: indication or conjecture, not yet fully proved.

Doc. 159 shows that the spectral zeta function of the Laplace operator on a torus T^d is the Epstein zeta function, whose one-dimensional special case is the Riemann zeta $\zeta(s) = \sum_{n=1}^{\infty} n^{-s}$. Doc. 342 shows that primes appear in the $\text{GF}(3^k)$ hierarchy of the $T^4/\mathbb{Z}_3$ orbifold ordered by entry depth $k = \text{ord}_p(3)$. This document combines both: what does the $\text{GF}(3^k)$ hierarchy imply for the zeta function itself?

.2 Theorem A: Orbifold projection and Dirichlet L-function

Decomposition of the zeta by $\mathbb{Z}_3$ classes

On $T^4/\mathbb{Z}_3$ the allowed modes $n \in \mathbb{Z}^4$ are structured by $\mathbb{Z}_3$ invariance. In the one-dimensional analogue $\zeta(s)$ splits into three partial sums by $n \bmod 3$ **[B]**:

$$\zeta(s) = \underbrace{\sum_{n=1}^{\infty} \frac{1}{n^s}}_{\substack{3|n \\ =3^{-s}\zeta(s)}} + \underbrace{\sum_{\substack{n=1 \\ n \equiv 1(3)}}^{\infty} \frac{1}{n^s}}_{L_1(s)} + \underbrace{\sum_{\substack{n=1 \\ n \equiv 2(3)}}^{\infty} \frac{1}{n^s}}_{L_2(s)} \quad (1)$$

The fixed-point sector ($n \equiv 0$) contributes exactly $3^{-s}\zeta(s)$; the twist sectors ($n \equiv \pm 1$) together $(1 - 3^{-s})\zeta(s)$.

Principal character $\chi_0 \bmod 3$

The Dirichlet character $\chi_0 \bmod 3$ with $\chi_0(n) = 0$ if $3 \mid n$, otherwise $\chi_0(n) = 1$, gives **[B]**:

$$L(s, \chi_0) = \sum_{\gcd(n,3)=1} \frac{1}{n^s} = (1 - 3^{-s})\zeta(s) = \prod_{p \neq 3} \frac{1}{1 - p^{-s}} \quad (2)$$

This is the **spectral zeta function of the $T^4/\mathbb{Z}_3$ torus**: the characteristic prime $p = 3$ does not appear as an Euler factor — it is the geometric foundation of the orbifold and stands outside the prime hierarchy.

Non-trivial character $\chi_1 \bmod 3$

The character $\chi_1 \bmod 3$ with $\chi_1(1) = +1$, $\chi_1(2) = -1$, $\chi_1(0) = 0$ measures the *asymmetry* of the twist sectors **[B]**:

$$L(s, \chi_1) = \sum_{\substack{n=1 \\ n \equiv 1(3)}}^{\infty} \frac{1}{n^s} - \sum_{\substack{n=1 \\ n \equiv 2(3)}}^{\infty} \frac{1}{n^s} \quad (3)$$

with the exact value (Dirichlet 1837): $L(1, \chi_1) = \pi/(3\sqrt{3})$.

Physical meaning **[S]**: $L(s, \chi_1)$ measures the difference between winding numbers $w = +1$ and $w = -1$ on the torus, i.e. the $\mathbb{Z}_3$ sector asymmetry. Whether $L(s, \chi_1)$ has a direct physical observable in FFGFT (particle–antiparticle asymmetry in the twisted sector?) remains open **[S]**.

.3 Theorem B: Hierarchical Euler product by $k = \text{ord}_p(3)$

The primes in $L(s, \chi_0) = \prod_{p \neq 3} (1 - p^{-s})^{-1}$ are not equivalent. By the entry-depth rule $k = \text{ord}_p(3)$ from Doc. 342, Theorem B, the Euler product is hierarchically ordered **[B]**:

$$L(s, \chi_0) = \prod_{k=1}^{\infty} \prod_{\substack{p \neq 3 \\ \text{ord}_p(3)=k}} \frac{1}{1 - p^{-s}} \quad (4)$$

Table 1: Hierarchy of Euler factors by $k = \text{ord}_p(3)$

k	new prime(s)	Euler factor at $s = 2$	harmonics	FFGFT role
1	2	$4/3 = 1.333$	octave	Z_2 fixed point, massive sector
3	13	≈ 1.006	thirteenth	GF(27)*, leptons
4	5	≈ 1.042	major third	factor in $(m_\mu/m_e)^2$
5	11	≈ 1.008	eleventh	neutrinos (Doc. 340)
6	7	≈ 1.021	seventh	top quark (Doc. 289)
≥ 7	$\geq 41, 1093, \dots$	$\approx 1 + p^{-2}$	no small harmonics	suppressed

Fractal suppression of higher Euler factors

On the fractal T^4 torus (Doc. 159) high-mode contributions are suppressed by the damping factor $\tau^{-(1+\xi)k}$. This transfers to the Euler factors; the *weighted* Euler product reads

$$\zeta_{T^4/Z_3}^{\text{frac}}(s) \cong \prod_{k=1}^{\infty} \prod_{\substack{p \neq 3 \\ \text{ord}_p(3)=k}} \left(\frac{1}{1 - p^{-s}} \right)^{\tau^{-k(1+\xi)}} \quad (5)$$

Since $\tau^{-k(1+\xi)} \rightarrow 0$ as $k \rightarrow \infty$, each Euler factor is pushed toward 1 for large k : primes with $\text{ord}_p(3) \geq 7$ make no physical contribution **[B]**. This is Theorem D of Doc. 342 in the language of the zeta function.

.4 Theorem C: New zeros from the orbifold projection

Zeros of $(1 - 3^{-s})$

The factor $(1 - 3^{-s})$ in $L(s, \chi_0) = (1 - 3^{-s})\zeta(s)$ has zeros at **[B]**:

$$3^{-s} = 1 \iff s = \frac{2\pi i k}{\ln 3}, \quad k \in \mathbb{Z} \setminus \{0\} \quad (6)$$

These lie on $\text{Re}(s) = 0$ (imaginary axis), *not* on the critical line $\text{Re}(s) = 1/2$ of the Riemann hypothesis. They are absent from the Riemann zeta spectrum; they arise solely from the Z_3 orbifold projection.

The zero spacing $2\pi/\ln 3 \approx 5.72$ encodes the periodicity of the Z_3 lattice structure. A resonance test of the first eight known Riemann zeros γ_n against this unit shows no significant coincidence (deviations 0.15–0.47) **[B]**: the full Riemann zeta encodes the distribution of *all* primes, not only the Z_3 -compatible ones.

Complete zero structure of the orbifold zeta

$L(s, \chi_0)$ has three types of zeros:

- **Trivial zeros of $\zeta(s)$:** at $s = -2, -4, -6, \dots$
- **Non-trivial Riemann zeros:** on $\text{Re}(s) = 1/2$ (assumed), unchanged
- **Orbifold zeros:** at $s = 2\pi i k / \ln 3$, $k \neq 0$ — new, generated by the $\mathbb{Z}_3$ projection **[B]**

.5 Theorem D: The four classes under sector pairing

The Galois character does not distinguish the classes

The Galois group $\text{Gal}(\text{GF}(27)/\text{GF}(3)) = \langle F \rangle \cong \mathbb{Z}_3$ with $F: x \mapsto x^3$ has three characters $\chi_j(F) = \omega^j$, $j = 0, 1, 2$. All four primitive classes $f_1, \dots, f_4$ of Doc. 342 are *regular* F -orbits of length 3: F acts on each as a 3-cycle, and the associated character is the same for all four. The Galois character therefore does not separate the classes **[B]**.

The distinction lies in the discrete logarithm: with a generator x of $\text{GF}(27)^*$, each class is an orbit $\{x^k, x^{3k}, x^{9k}\}$ with $\gcd(k, 26) = 1$, i.e. a coset of $\langle 3 \rangle = \{1, 3, 9\}$ in $(\mathbb{Z}/26)^* \cong \mathbb{Z}_{12}$. There are $12/3 = 4$ cosets **[B]**:

Table 2: The four primitive classes as cosets of $\langle 3 \rangle$ in $(\mathbb{Z}/26)^*$ (basis $x^3 + 2x + 1 = 0$)

Class	Minimal polynomial	Exponents k	Role
f_2	$x^3 + 2x + 1$	$\{1, 3, 9\}$	Doc. 341, lepton layer
f_3	$x^3 + 2x^2 + 1$	$\{17, 23, 25\}$	$= f_2^{-1}$
f_1	$x^3 + 2x^2 + x + 1$	$\{5, 15, 19\}$	
f_4	$x^3 + x^2 + 2x + 1$	$\{7, 11, 21\}$	$= f_1^{-1}$

The quotient $(\mathbb{Z}/26)^* / \langle 3 \rangle \cong \mathbb{Z}_4$ is cyclic; inversion $k \mapsto -k$ is its element of order 2.

Sector pairing reduces four to two

Doc. 336 identifies the FFGFT sector pairing $k \leftrightarrow -k$ with the Frobenius automorphism; at the level of elements of $\text{GF}(27)^*$ it is inversion $x \mapsto x^{-1} = x^{25}$. It permutes the classes as **[B]**

$$f_1 \leftrightarrow f_4, \quad f_2 \leftrightarrow f_3. \quad (7)$$

If sector pairing is a symmetry of the mass layer — as it is in the corpus — there are not four but **two** physically distinguishable $\text{GF}(27)^*$ layers: $\{f_2, f_3\}$ (leptons, Doc. 341) and $\{f_1, f_4\}$.

Negation couples order 26 to order 13

Since $-1 = x^{13}$ in $\text{GF}(27)$, the map $x \mapsto -x$ sends every element of order 26 to one of order 13 and vice versa. The four order-13 classes $g_1, \dots, g_4$ of Doc. 342 are therefore not an independent structure but the sign partners of the primitive classes **[B]**:

$$f_1 \leftrightarrow g_1, \quad f_2 \leftrightarrow g_3, \quad f_3 \leftrightarrow g_2, \quad f_4 \leftrightarrow g_4. \quad (8)$$

The candidate “sub-lepton structure” from the order-13 polynomials, left open in Doc. 342, is thereby eliminated.

Open question

The reduction $4 \rightarrow 2$ sharpens the question of Doc. 342: not “do f_2, f_3, f_4 carry further layers”, but “is there exactly one second $\text{GF}(27)^*$ layer $\{f_1, f_4\}$ besides the lepton layer $\{f_2, f_3\}$, and which particles carry it?” This remains open **[S]**.

.6 Summary

Table 3: Results and status

Theorem	Content	Status
A	$\zeta_{T^4/\mathbb{Z}_3}(s) = (1 - 3^{-s})\zeta(s) = L(s, \chi_0)$	[B]
A'	$L(s, \chi_1)$ measures twist-sector asymmetry	[S]
B	Euler product hierarchical by $k = \text{ord}_p(3)$	[B]
B'	Fractal damping suppresses Euler factors $k \geq 7$	[B]
C	Orbifold zeros at $s = 2\pi ik / \ln 3$ on $\text{Re}(s) = 0$	[B]
C'	No resonance of Riemann zeros with $\mathbb{Z}_3$ unit	[B]
D	Galois character does not separate; classes $= (\mathbb{Z}/26)^* / \langle 3 \rangle \cong \mathbb{Z}_4$	[B]
D'	Sector pairing $k \leftrightarrow -k$: $f_1 \leftrightarrow f_4, f_2 \leftrightarrow f_3$ — 4 classes $\rightarrow 2$	[B]
D''	Negation couples order 26 to 13; no sub-lepton layer	[B]
D'''	Second layer $\{f_1, f_4\}$ physically realised?	[S]

The central statement: the full Riemann zeta $\zeta(s)$ belongs to the flat $\mathbb{R}^4$ spectrum. The zeta of the $T^4/\mathbb{Z}_3$ orbifold is $L(s, \chi_0) = (1 - 3^{-s})\zeta(s)$ — a filtered version in which the 3-Euler factor is separated out and the primes are weighted by the Galois hierarchy $k = \text{ord}_p(3)$. This is the zeta-function form of the entry rule of Doc. 342.

Check scripts

Directory 2/python/Dok343_Skripte/, wrapper `run_all_343.sh`:

- `pruef_343_zeta_galois.py` — Theorems A–D; partial zeta decomposition, L-functions, zeros, resonance test, weighted Euler product.
- `pruef_343b_klassen_symmetrie.py` — Theorem D; explicit Frobenius classes in $\text{GF}(27)^*$, action of inversion and negation, cosets of $\langle 3 \rangle$ in $(\mathbb{Z}/26)^*$, exact values of $L(s, \chi_1)$.

All assertions passed.

Note on the use of AI assistance

This document was prepared with the assistance of Claude (Anthropic). All numerical computations are verified by Python check scripts. The physical interpretations and conclusions are those of Johann Pascher (ORCID 0009-0000-6518-4064).

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